

SecureChannel Project - Work Distribution

Team Members

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`crypto/sha256.py`

Implement SHA-256 from the official specification and verify it using the provided test vectors.

`crypto/hmac.py`

Implement HMAC-SHA256 using our SHA-256 implementation and use it for handshake authentication.

`crypto/hkdf.py`

Implement HKDF (Extract and Expand) and use it to derive session keys from the shared secret while incorporating the PSK as the salt.

`protocol/transcript.py`

Build the handshake transcript containing both public keys, identities, and protocol information.

`protocol/key_schedule.py`

Derive the client and server session keys and generate the initial nonce material.

protocol/message.py

Define the message structure used by both parties, including headers, sequence numbers, and associated data fields.

network/client.py

Implement the client side of the application, including connection establishment, handshake execution, and secure message exchange.

bonus/gui.py

Implement a simple graphical interface for connecting, sending messages, and displaying received messages.

tests/test_sha256.py

Validate SHA-256 against official test vectors.

tests/test_hmac.py

Validate HMAC against official test vectors.

tests/test_hkdf.py

Validate HKDF against official test vectors.

Documentation

Prepare the README file and document the hashing, authentication, key derivation, and GUI components in the final report.

Amjad Qaher

crypto/chacha20.py

Implement the ChaCha20 stream cipher and verify it using official test vectors.

crypto/poly1305.py

Implement Poly1305 and verify it using official test vectors.

crypto/aead.py

Implement the ChaCha20-Poly1305 AEAD construction, including encryption, authentication, and associated data processing.

crypto/x25519.py

Implement X25519 key exchange and verify it using official test vectors.

protocol/handshake.py

Implement the handshake process, including key exchange and authentication.

protocol/replay.py

Implement replay protection using sequence numbers and message validation.

network/server.py

Implement the server side of the application, including accepting connections, performing the handshake, and exchanging secure messages.

tests/test_chacha20.py

Validate ChaCha20 against official test vectors.

tests/test_poly1305.py

Validate Poly1305 against official test vectors.

tests/test_aead.py

Validate ChaCha20-Poly1305 against official test vectors.

tests/test_x25519.py

Validate X25519 against official test vectors.

Documentation

Document the encryption, authenticated encryption, key exchange, handshake, and replay-protection components in the final report.

Shared Responsibilities

- Integrate all project components.
- Verify that all official test vectors pass successfully.
- Review each other's code and understand the complete system.
- Prepare the final report, references section, AI usage statement, and individual contribution statements.
- Prepare for the oral discussion and be able to explain the full protocol flow from key exchange to secure messaging.